Robustness of FNETS: Model Performance Under Violation of Assumptions

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Abstract

This paper empirically evaluates the robustness of FNETS, a factor-adjusted network estimation and forecasting method for high-dimensional time series, under violation of its core assumptions. Applying FNETS to a panel of 46 U.S. financial firms, we compare model behavior across a calm period (2004-2006) and the Great Financial Crisis (2007-2009). Our result reveals that the stationarity assumption fails almost entirely during the crisis, and estimated network density rises substantially, consistent with correlated deleveraging across the financial sector. Counterintuitively, maximum forecast error improves significantly during the crisis despite these violations, a forecasting paradox we attribute to the sustained co-movement of firm volatility under common systemic exposure. Our results suggest that FNETS retains practical forecasting utility even when its theoretical assumptions are severely challenged.

1 Introduction

The 2008 Great Financial Crisis exposed the endogenous vulnerability of the global financial system, especially the U.S. one, and the speed at which volatility shocks can spread across institutions based on more and more inter-correlations brought by new financial technologies. As early as February 2007, subprime mortgage default rates began rising sharply and house prices fell for the first time in decades; by August 2007, repo market haircuts surged and the market for short-term asset-backed commercial paper collapsed; and by September 2008, the failure of Lehman Brothers triggered system-wide contagion through contractual interdependence, fire sales, and adverse selection across financial intermediaries (Brunnermeier, 2009).

Modeling these transmission channels requires methods capable of handling high-dimensional panels of financial time series exhibiting strong serial and cross-sectional dependence. Barigozzi, Cho, and Owens (2024) propose a factor-adjusted network estimation (FNETS) which is a forecasting methodology that addresses closely to this challenge. FNETS decomposes each series into a common factor-driven component and an idiosyncratic component modeled as a sparse VAR, enabling the recovery of Granger causal, contemporaneous, and long-run partial correlation networks after removing pervasive market-wide co-movement. The methodology rests on four core assumptions: sufficiently strong common factors, algebraic decay of factor filters, causality and sparsity of the idiosyncratic VAR, and finite higher-order moments permitting fat tails. Although the authors established rigorous theories under these assumptions, it had not been formalized to assess whether that they hold in empirical financial data.

This paper addresses this gap. We apply FNETS to a panel of 46 U.S. financial firms and evaluate its assumptions and performance across two sharply contrasting market regimes around the financial crisis. The first is a calm period that spanned 2004-2006, during which credit markets expanded steadily, securitization volumes surged, and financial volatility remained subdued. The S&P / Case-

Shiller Home Price Index continued rising throughout this period and mortgage delinquency rates reached historic lows, reflecting an environment in which cross-firm dependence was moderate and volatility was mean-reverting, that is, the conditions under which FNETS assumptions are expected to hold (Brunnermeier, 2009; Adelino, Schoar and Severino, 2018).

The second is a crisis period that occurred from 2007 to 2009. The onset was marked by a sharp rise in subprime mortgage defaults in February 2007, the collapse of repo markets and asset-backed commercial paper in August 2007, and the failure of Lehman Brothers in September 2008, which triggered a fast-speed system-wide contagion (Brunnermeier, 2009). The NBER formally defines December 2007 to June 2009 as the Great Recession, a period characterized by extreme volatility persistence, near-universal non-stationarity in financial time series, and sharply elevated cross-firm interconnectedness which are conditions that directly challenge the core assumptions of FNETS. Together, these two periods provide a natural laboratory for stress-testing the model under conditions ranging from assumption-consistent to assumption-violating.

Based on the setting described above, our analysis makes two contributions. First, we systematically test each empirically relevant FNETS assumption across both regimes and document how assumption validity can deteriorate under a crisis. Second, we estimate FNETS networks and evaluate forecasting performance in both periods, and document changes in estimated network structure.

2 Model: FNETS

Directly applying VAR models to high-dimensional panels is infeasible, as the number of parameters grows quadratically with the cross-sectional dimension p. Furthermore, financial panels typically exhibit strong serial and cross-sectional correlations driven by common factors, which standard VAR estimators fail to account for (Barigozzi et al., 2024). FNETS addresses this by first removing the dominant factor-driven co-movements via dynamic PCA, and then modeling the remaining idiosyncratic dependence as a sparse VAR process, whose nonzero entries define a Granger causality network among the series.

Barigozzi et al. (2024) model a zero-mean, second-order stationary p-variate process $X_t = (X_{1t}, ..., X_{pt})'$ as the sum of two latent components:

$$X_t = \chi_t + \xi_t$$

where χ_t is a factor-driven common component and ξ_t is an idiosyncratic component. The common component is specified as:

$$\chi_t = B(L)u_t = \sum_{l=0}^{\infty} B_l u_{t-l}$$

where $u_t = (u_{1t}, ..., u_{qt})'$ is a q-dimensional vector of common factors with $E(u_t) = 0$ and $cov(u_t) = I_q$. The idiosyncratic component follows a VAR(d) process:

$$A(L)\xi_t = \xi_t - \sum_{l=1}^{d} A_l \xi_{t-l} = \Gamma^{1/2}\varepsilon_t$$

where $\varepsilon_t = (\varepsilon_{1t}, ..., \varepsilon_{pt})'$ satisfied $E(\varepsilon_t) = 0$ and $cov(\varepsilon_t) = I_p$ and $\Gamma \in R^{p\times p}$ is a positive definite matrix.

The VAR coefficient matrices $A_1, ..., A_d$ are stacked into $\beta = [A_1, ..., A_d]' \in R^{pd\times p}$, so that the idiosyncratic VAR can be written compactly as $\xi_t = \beta' Z_{t-1} + \Gamma^{1/2}\varepsilon_t$ where $Z_{t-1} = (\xi'_{t-1}, ..., \xi'_{t-d})$. The sparsity of β directly determines the sparsity of the estimated Granger causality network, as a directed edge (i, i') exists if and only if $\hat{\beta}_{l,ii'} \neq 0$ for some lag l.

2.1 Assumptions

FNETS model was stated to be mainly based on the following four assumptions.

Assumption 1 Common Factor Strength

The j-th dynamic eigenvalue of the spectral density matrix of χ_t diverges at rate p^{ρ^j} as $p \longrightarrow \infty$, where $\rho_j \in (3/4, 1]$ for all $j = 1, ..., q$. This ensures that each factor is sufficiently strong to be identified and separated from the idiosyncratic component via dynamic PCA. The case ($\rho_j = 1$) corresponds to fully pervasive strong factors, while $\rho_j < 1$ permits the presence of weak factors. The requirement that $\rho_j > 3/4$ is a minimal one (Barigozzi et al., 2024). That is, Assumption 2.1 requires that the common factors, such as market-wide shocks or sector-level co-movements, exert a sufficiently strong and pervasive influence across all p series. If this condition fails, or the common factors are too weak, FNETS' dynamic PCA cannot reliably separate the common component from the idiosyncratic component, rendering the subsequent network estimation invalid. In the context of our 46 U.S. financial firms, Assumption 2.1 is likely satisfied by construction: given that all series belong to the same nation and same sector, they are theoretically expecting to have strong common factors that driven by shared regulatory environment, interest rate exposure, and correlated business cycles, etc., to dominate the cross-sectional variation. Therefore, this assumption is not going to be tested in the following.

Assumption 2 Algebraic Decay of Factor Filters

Assumption 2 is imposed, together with Assumption 3, to control the serial dependence in X_t.

Specifically, it requires both the row and column wise 12 norms of the filter coefficient matrices B_l to decay at an algebraic rate of order $(1+l)^{-\varsigma}$ for some $\varsigma > 2$ (Barigozzi et al., 2024). In our view, this is more a technical regularity condition other than an empirical one, thus it is not going to be tested in our analysis.

Assumption 3 Causality and Sparsity of Idiosyncratic VAR

This assumption imposes causality and sparsity conditions on the idiosyncratic VAR process ξ_t, including standard sub-conditions in the literature (Lutkepohl, 2005). Specifically, one requires that d is a finite positive integer and

$$det(A(z)) \neq 0, \forall |z| \leq 1$$

which ensures that ξ_t is causal, together with another condition guarantees finite and nonzero covariance of xi_t via $m_\varepsilon I \leq \Gamma \leq M_\varepsilon I$ (Barigozzi et al. 2024).

The third sub-condition requires the smallest dynamic eigenvalue of the spectral density of ξ_t to be uniformly bounded away from zero:

$$\inf_{\omega \in [-\pi, \pi]} \mu_{\xi,p}(\omega) \geq m_\xi > 0$$

.

As noted in the paper, a sufficient condition for this is that the row and column l_1 norms of the VAR coefficient matrices are uniformly bounded, which can be interpreted as requiring that the VAR structure is sufficiently sparse. A fourth sub-condition, as stated in original paper's assumption 2.3(iv), imposes a similar algebraic decay condition on the Wold representation coefficients $D_{l,ik}$ of ξ_t, which together with Assumption 2 ensures that the serial dependence in the full observed process X_t decays at an algebraic rate (Barigozzi et al., 2024). The intuition is that after removing the dominant common component via factor adjustment, the remaining idiosyncratic dependence is expected to be weak, making sparse VAR modeling a reasonable approximation.

Since the causality condition implies stationarity (Lutkepohl, 2005), stationarity is a necessary condition for causality. We therefore use the Augmented Dickey-Fuller (ADF) test applying it to each series individually as an indirect diagnostic because failure to reject the unit root null provides evidence against stationarity and, by necessity, against the causality condition as well. We assessing sparsity descriptively following the sparsity measure defined in the paper. Specifically, Barigozzi et al. (2024) measures sparsity of the VAR coefficient matrix β via the total number of nonzero entries

$$s_0 = \sum_{j=1}^{p} s_{0,j}$$

and the maximum in-dgree

$$s_{in} = \max_{1 \leq j \leq p} s_{0,j}$$

, where $s_{0,j} = |\beta_{\cdot j}|_0$ denotes the number of nonzero coefficients in column j, corresponding to the number of incoming edges at node j in the Granger network N^G. We report the overall network edge density and the ratio s_{in}/p as descriptive indicators of whether the sparsity condition is plausibly satisfied in each sub-period.

Assumption 4 Moment conditions

Barigozzi et al. (2024) use this assumption to govern the probabilistic properties of the innovations driving both the common and idiosyncratic components. The first three sub-conditions impose standard martingale difference structure and orthogonality between the two innovation processes. The key empirically relevant condition requires that innovations possess finite moments up to some order $\nu > 4$ and

$$\max(\max_{1\leq j\leq q} E(|u_{jt}|^{\nu}), \max_{1\leq i\leq p} E(|\varepsilon_{it}|^{\nu})) \leq \mu_{\nu}$$

.

This moment condition excludes extremely heavy-tailed distributions for which higher order moments fail to exist. However, comparing to Barigozzi & Brownlees (2019), Barigozzi et al. (2024) relaxed the rail condition from requiring light-tailed distributions to only requiring finite moments of order $\nu > 4$, permitting a wide range of fat-tailed distributions. We assess whether this condition is plausibly satisfied in each sub-period using series-wise kurtosis, the Jarque-Bera test, and the Anscombe-Glynn test for excess kurtosis.

2.2 FNETS Estimation Procedures

FNETS estimation proceeds in three steps. In step 1, the common component χ_t is estimated via dynamic principal component analysis (dynamic PCA) (Forni et al., 2000) on the spectral density matrix of X_t, estimated using Bartlett kernel (Priestley, 1982) with bandwidth $m = \lfloor 4(\frac{n}{\log n})^{(\frac{1}{3})} \rfloor$, and the q largest dynamic eigenvalues and their associated eigenvectors are retained to recover the auto-covariance structure of χ_t and ξ_t. The number of factors q is selected via the information criterion of Hallin and Liska (2007).

In step 2, the idiosyncratic VAR parameters are estimated via an l_1-regularized Yule-Walker estimator. Letting G and g denote matrices constructed from the estimated auto-covariances of ξ_t, the coefficient matrix β is estimated as

$$\hat{\beta} = \arg \min_{M \in R^{pd \times p}} tr(MGM' - 2Mg) + \lambda ||M||_1$$

where $\lambda > 0$ is a tuning parameter selected via cross-validation. Let $V = \{1, ..., p\}$ denote the vertex set corresponding to the p time series. The Granger causality network is defined as the directed graph $N^G = (V, E^G)$, where the edge set is given by

$$E^G = \{(i, i') \in V \times V : \hat{\beta}_{l,ii'} \neq 0 \text{ for some } 1 \leq l \leq d\}$$

,

so that a directed edge $(i, i') \in E^G$ indicates that the idiosyncratic component of series i' Granger causes that of series i at some lag l (Barigozzi et al., 2024; Barigozzi and Brownlees, 2019). The sparsity of $\hat{\beta}$ therefore determines the sparsity of the estimated network, and the assumption that ξ_t admits a sparse VAR representation is equivalent to assuming that N^G has few edges relative to the p^2 possible directed pairs.

In step 3, one-day-ahead forecasts are constructed as:

$$\hat{X}_{T+1|T} = \bar{X} + \hat{\chi}_{T+1|T} + \hat{\xi}_{T+1|T}$$

where $\hat{\chi}_{T+1|T}$ is the forecast of the common component under the restricted static factor model, and $\hat{\xi}_{T+1|T}$ is the VAR-based forecast of the idiosyncratic component. Forecasting performance is evaluated using two metrics averaged over all firms at each trading day T as following,

$$FE_{avg} = \frac{\sum_{i=1}^{p} (X_{i,T+1} - \hat{X}_{i,T+1|T})^2}{\sum_{i=1}^{p} X_{i,T+1}^2}, FE_{max} = \frac{\max_i |X_{i,T+1} - \hat{X}_{i,T+1|T}|}{\max_i |X_{i,T+1}|}$$

3 Empirical Study

This paper empirically evaluates the robustness of FNETS by examining whether its core assumptions hold under different market conditions, and assessing the extent to which assumption violations affect model performance.

Our analysis proceeds in two steps. First, we empirically test the key assumptions of FNETS within each subperiod, providing evidence on which assumptions are satisfied and which are violated, and to what degree. Second, we evaluate the out-of-sample forecasting performance of FNETS under a rolling window framework, and formally test whether the difference in performance across regimes is statistically significant. Because neither Barigozzi et al. (2024) nor Barigozzi and Brownlees (2019) conduct a systematic assumption test on their empirical datasets, our analysis fills this gap by directly linking the degree of assumption violation to observed model behavior across market regimes.

3.1 Data

We use the dataset from Barigozzi et al. (2024) directly. The panel consists of $p = 46$ U.S. publicly listed companies, all classified as financial under the Global Industry Classification Standard, with stock prices retrieved from the Wharton Research Data Service. The full sample spans January 3, 2000 to December 31, 2012, covering 3267 trading days. According to Barigozzi et al. (2024), they followed Diebold and Yilmaz (2014) and measured volatility using the high-low range estimator $\sigma_{it}^2 = 0.361(p_{it}^{high} - p_{it}^{low})^2$, where p_{it}^{high} and p_{it}^{low} denote the daily maximum and minimum log-price of stock i, and the series entering the model are the log-transformed volatility measures $X_{it} = log(\sigma_{it}^2)$

To examine whether the core assumptions of FNETS hold under different market conditions, we partition the sample into two sub-periods: a calm period spanning 2004-2006, and a crisis period spanning 2007-2009 corresponding to the Great Financial Crisis.

3.2 Methods

3.2.1 Assumption Tests Procedures

As we mention in section 2.1 Assumptions, we are not going to test Assumption 1 Common Factor Strength as all 46 firms belong to the same nation and sector, strong common factors are expected by construction, and the assumption is not empirically tested, and Assumption 2 Algebraic Decay of Factor Filter as it is is a technical regularity condition rather than an empirically testable one. We are going to test Assumption 3 Causality and Sparsity, and Assumption 4 Heavy tails.

Stationarity

For Assumption 3 Causality and Sparsity of Idiosyncratic VAR, first, we are going to use Augmented Dickey-Fuller (ADF) test (Said and Dickey, 1984; Banerjee et al., 1993) to assess the **stationarity** of both calm and crisis periods to approximately measure the causality of idiosyncratic VAR. ADF test is applied to each of the 46 series individually using the ADF test function from the R package **tseries** (Trapletti & Hornik, 2026). For each series, the null hypothesis is the presence of a unit root against the stationary alternative. The lag length is set to the default value of $\lfloor (T-1)^{\frac{1}{3}} \rfloor$, and the procedure is repeated across all 46 firms, yielding 46 individual p-values from which we compute the proportion of series rejecting the unit root null at the 5% significance level.

Sparsity

Second, sparsity is assessed descriptively following the measures defined in Barigozzi et al. (2024).

Within each rolling window of both length $n = 252$ and $n = 126$, we estimate the idiosyncratic VAR(1) via l_1-regularized Yule-Walker as implemented in FNETS, and extract the estimated coefficient matrix $\hat{\beta}$. Then, we compute the network edge density $\frac{\hat{s}_0}{p^2}$ where $\hat{s}_0 = ||\hat{\beta}||_0$ is the total number of nonzero entries, and the maximum in-degree $\hat{s}_{in} = \max_j ||\hat{\beta}_{\cdot j}||_0$. These quantities are averaged over all rolling windows within each sub-period and compared across the calm and crisis periods.

Heavy Tails

We assess the *Assumption 4 Moment conditions* using three complementary series-level statistics, each applied to all 46 firms individually. The three measures are chosen to distinguish between different sources of non-normality, which is important for interpreting the assumption in the context of financial crisis data.

First, we compute the sample kurtosis $\hat{K}_i = \frac{\hat{\mu}_4}{\hat{\mu}_2^2}$ for each series and report its cross-sectional mean as a direct descriptive indicator of tail behavior. This measure quantifies the magnitude of the fourth standardized moment but provides no formal statistical inference.

Second we apply the Jarque-Bera test from the R packages **tseries** (Cromwell et al., 1994; Said & Trapletti & Hornik, 2026) to each series. This is a joint test using both the skewness and kurtosis coefficients, with the null hypothesis of normality. Rejection indicates departure from normality in either or both the third or fourth standardized moment, but not specifying which.

Third, we apply the two-sided alternative Anscombe-Glynn test of kurtosis from the R package **moments** (Anscombe and Glynn, 1983; Komsta & Novomestky, 2022) to each series. Under the null hypothesis of normality, kurtosis equals 3. The rejection of this hypothesis indicates that kurtosis is significantly different from 3 and provides inference specifically on tail behavior which is most directly relevant to Assumption 4.

By combining all three measures, we are able to disentangle whether any observed non-normality during a given sub-period is driven primarily by heavy tails or by skewness. We report the proportion of the 46 series rejecting each null at the 5% significance level, alongside the cross-sectional mean kurtosis.

3.2.2 Implementation Details

Our implementation follows the FNETS methodology of Barigozzi et al. (2024) with several simplifications tailored to the scope of this paper. First, we exclude the FARM forecasting method (Fan et al., 2021), which serves as a baseline comparison in the original paper, as our focus is on evaluating FNETS itself rather than benchmarking against alternative methods. Second, we omit the Dantzig selector variant of FNETS ($\hat{\beta}^D$), which is reported in the supplementary appendix of the original paper as a robustness check. Third, we exclude the factor-only forecast component ($\hat{\chi}_{T+1|T}$) and assess only the full LASSO-based FNETS forecast, which combines both the common and idiosyncratic components, as this reflects the overall predictive performance of the model. Fourth, we use only the restricted static factor model for forecasting the common component, as the unrestricted GDFM-based forecast is not the primary method of the original paper. Finally, we fix the VAR lag at $d = 1$ throughout, rather than selecting over $d \in \{1, ..., 5\}$ as in the original sensitivity analysis, since our focus is on assumption testing and its effects on FNETS forecast outputs rather than VAR lag selection.

3.2.3 Statistical Inference and Robustness Check

To formally assess whether FNETS forecasting performance differs significantly across the two sub-periods, we apply Welch two-sample t-tests to the rolling-window forecasting errors with default significant level $\alpha = 0.05$. Specifically, for each of the two error metrics FE_{avg} and FE_{max}, we test the null hypothesis that the mean forecasting error is equal across the calm and crisis periods against a two-sided alternative. The Welch variant is used in place of the standard two-sample t-test as it does not assume equal variances across the two groups, which is appropriate given that the calm and crisis periods may differ substantially in their error distributions.

As a robustness check, we repeat the full forecasting and sparsity diagnostic procedure using a shorter rolling window of $n = 126$ trading days, corresponding to approximately half a calendar year. First, it assesses whether our main findings under $n = 252$ are sensitive to the choice of window length. Second, it carries a natural economic interpretation, that is, a shorter window imposes a weaker assumption of structural stability within the estimation period, which may be more appropriate during the crisis period when the dependence structure among financial firms evolves rapidly. The Welch t-tests are subsequently applied to the $n = 126$ results as well, enabling a direct comparison of statistical significance across both window lengths. Note that the stationarity and heavy tail assumption tests are not repeated under $n = 126$, as these are properties of the underlying data and are independent of the rolling window length.

4 Results

4.1 Assumption Tests

4.1.1 Stationarity & Heavy Tail

	Calm (2004-2006)	Crisis (2007-2009)
Stationarity (ADF, $p < 0.05$)	100%	4.3%
ADF p-value range	[0.01, 0.01]	[0.0383, 0.7162]
Mean kurtosis	3.392	2.838
JB reject normality	63.0%	76.1%
AG reject kurtosis = 3	34.8%	41.3%

Table 1 reports the stationarity and heavy tail test results for both the calm (2004-2006) and crisis (2007-2009) subperiods. For stationarity, during the calm period, all 46 vAR series reject the null hypothesis, which states as not stationary, under the Augmented Dickey-Fuller (ADF) test at the 5% significance level, with p-values uniformly at the minimum reportable value of 0.01. In contrast, during the crisis period, only 4.3% of the 46 VAR series reject the unit root null hypothesis, with p-values ranging from 0.038 to 0.716. (see Figure A3 in the Appendix for a visual overview of the mean log-volatility dynamics across the two sub-periods.)

For heavy tails test, sample kurtosis mean results as 3.392 during the calm period and 2.838 during the crisis period. Under the Jarque-Bera test, 63.0% of series reject the null hypothesis of not normality during the calm period, and it rising to 76.1% during the crisis period. Under the Anscombe-Glynn test against kurtosis equal to 3, 34.8% of series reject the null hypothesis during the calm period, compared to 41.3% during the crisis period.

4.1.2 Sparsity

	Calm (n=252)	Crisis (n=252)	Calm (n=126)	Crisis (n=126)
Edge density	0.1008	0.1316	0.1636	0.2528
s_{in}	11.06	17.34	15.10	23.75
q	1.15	1.60	1.01	1.14

Table 2 reports the sparsity measures averaged over the rolling windows within each sub-period. During the calm period, the mean edge density is 0.1008, the mean maximum in-degree $\bar{s}_{in}$ is 11.06, and the mean estimated number of factors $\bar{q}_1.15$ is 1.15. During the crisis period, these values increase to 0.1316, 17.34, and 1.60, respectively.

Although edge density rises from 0.1008 to 0.1316 during the crisis, the network remains sparse in absolute terms because 0.1316 corresponds to approximately 272 edges out of 2,070 possible directed pairs. The sparsity assumption is therefore not severely violated in the sense of network saturation, but rather the degree of sparsity deteriorates relative to the calm period, with maximum in-degree rising from 11.06 to 17.34 (approximately 38% of nodes).

Figure 1 plots the rolling-window (n = 252) edge density of the estimated idiosyncratic network over the sample period. Edge density remains relatively stable during the calm period (2004-2006), fluctuating around a mean of 0.1008. A remarkable increase is observed from 2007 onward, with edge density rising and remaining elevated throughout the crisis period (2007-2009), averaging 0.1316.

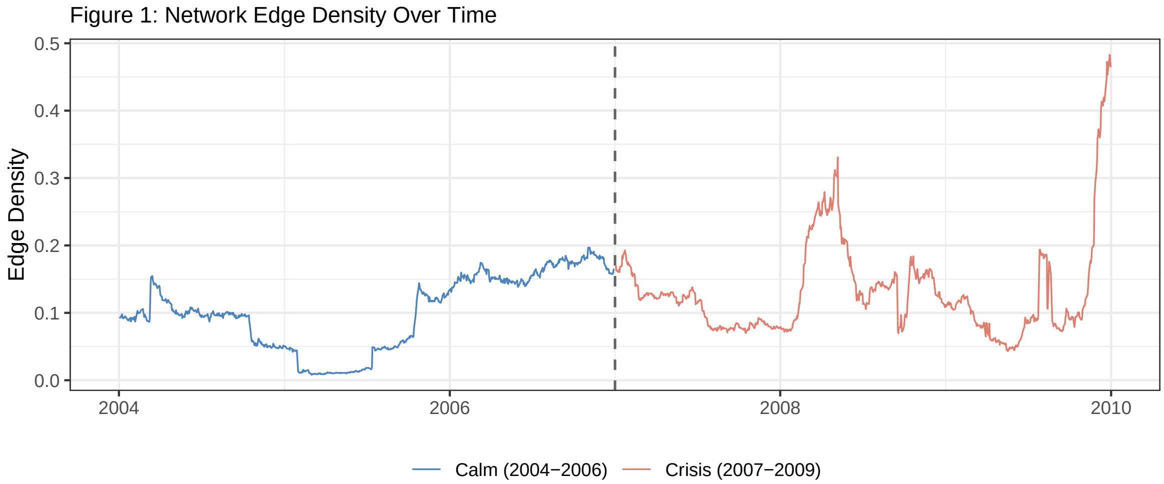

Figure 1: Network Edge Density Over Time

4.2 FNETS Forecasting Performance

	Calm (n=252)	Crisis (n=252)	Calm (n=126)	Crisis (n=126)
FE_{avg} mean	0.5884	0.5529	0.5944	0.5620
FE_{avg} median	0.4006	0.3827	0.3904	0.3806
FE_{max} mean	0.8526	0.7443	0.8552	0.7537
FE_{max} median	0.7858	0.7107	0.7864	0.7117

Table 3 reports the forecasting performance of FNETS across the two sub-periods under both rolling window specifications. For the main specification ($n = 252$), the mean FE_{avg} is 0.5884 during the calm period and 0.5529 during the crisis period. The mean FE_{max} is 0.8526 during the calm period and 0.7443 during the crisis period. The robustness check ($n = 126$) yields consistent results, with mean FE_{avg} of 0.5944 and 0.5620, and mean FE_{max} of 0.8552 and 0.7537, for the calm and crisis periods respectively.

Figure 2 plots the rolling-window FE_{max} over the sample period. During the calm period (2004-

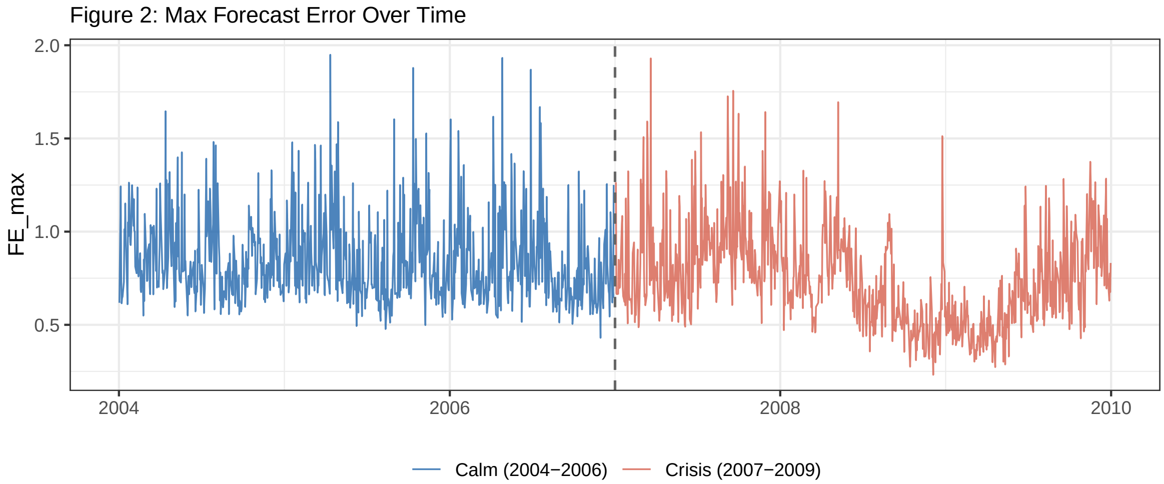

Figure 2: Max Forecast Error Over Time

2006), FE_{max} fluctuates at a relatively high level with considerable high variances. From 2007 onward, FE_{max} displays a sharp spike at the onset of the crisis, followed by a decline and stabilization at a lower level through the remainder of the crisis period (2007-2009).

	n = 252		n = 126	
	t-statistic	p-value	t-statistic	p-value
FE_{avg} mean	1.157	0.248	1.008	0.314
FE_{max} mean	8.394	< 0.001	7.584	< 0.001

Table 4 reports the Welch two-sample t-test results comparing forecast errors across the two sub-periods with significant level $\alpha = 0.05$. For FE_{avg}, the difference between the calm and crisis periods is not statistically significant under either n = 252 window with t-statistic $t = 1.157$, and corresponding p-value $p = 0.248$, or $n = 126$ window with t-statistic $t = 1.008$ and corresponding p-value $p = 0.314$. For FE_{max}, the difference is highly significant under both specifications. For rolling window $n = 252$, we observe a t-statistic $t = 8.394$ and the corresponding p-value $p < 0.001$; for rolling window $n = 126$, we observe a t-statistic $t = 7.584$ and the corresponding p-value $p < 0.001$.

5 Discussion

Our results reveal that, while the crisis period is characterized by systematic violations of Assumption 3, the forecasting performance of FNETS does not deteriorate accordingly, and in fact, may have a improve along the maximum error dimension.

5.1 Stationarity violation

The near-complete failure of stationarity during the crisis period, from the perspective of FNETS theory, suggests that the stationarity guarantee underpinning VAR estimation and network recovery is no longer credible during this period. While ADF non-rejection does not directly falsify the causality condition, it constitutes evidence against the necessary condition stationarity requires, and therefore casts doubt on the validity of Assumption 3 during the crisis period.

We interpret this not as evidence of genuine unit root non-stationarity, but rather as a reflection of extreme volatility persistence during the crisis. The shadow banking system, which functioned well against idiosyncratic risk in normal times, could not withstand the systemic risk that materialized from August 2007 onward, when the market for short-term asset-backed commercial paper collapsed and overnight lending in the shadow banking system proved far riskier than anticipated (Covitz et al., 2013). This system-wide shock subjected all 46 financial firms to correlated and persistent volatility, causing mean-reversion to effectively cease over the horizon relevant to the ADF test.

5.2 Rising interconnectedness during the crisis

The substantial increase in estimated network density during the crisis period is consistent with the system-wide funding shock discussed in Section 5.1: as the shadow banking system collapsed and financial institutions were simultaneously unable to roll over their short-term debt, correlated deleveraging drove firms toward more tightly coupled volatility dynamics (Brunnermeier, 2009; Covitz et al., 2013). It is worth noting, however, that the sparsity assumption is only partially stressed rather than fundamentally broken because the crisis-period network still comprises approximately 272 edges out of 2,070 possible directed pairs, remaining approximately sparse. The more severe violation lies elsewhere: this rise in estimated density occurs precisely when the stationarity assumption fails almost entirely, suggesting that the FNETS network estimates during the crisis should be interpreted with caution, as they may partially reflect non-stationary volatility dynamics rather than the true idiosyncratic Granger causal structure.

5.3 The Forecast Paradox

The significant improvement in FE_{max} during the crisis period, despite the severe violation of the stationarity assumption, is the most counterintuitive finding of this paper. The rising interconnectedness documented in Section 5.2 provides a natural starting point for interpretation. Since all 46 firms belong to the financial sector and shared a high degree of common exposure to the same systemic shock, their stock volatility likely moved together in a sustained directional pattern, making one-day-ahead forecasts structurally easier relative to the calm period when idiosyncratic variation dominated and no persistent trend existed to exploit. Nonetheless, we acknowledge that formally establishing this mechanism is beyond the scope of this paper, and leave a more rigorous investigation to future work.

5.4 Heavy Tail test & Robustness check

Regarding Assumption 4, the heavy tail results do not indicate a severe violation in either period, and are unlikely to be a primary driver of the observed differences in model performance. Notably, mean kurtosis declines from 3.392 in the calm period to 2.838 in the crisis period, falling below the normal benchmark of 3, while AG rejection rates simultaneously increase. This apparent contradiction is reconciled by the cross-sectional heterogeneity in kurtosis across the 46 firms: the distribution of firm-level kurtosis becomes more dispersed during the crisis, with some firms exhibiting kurtosis well below 3 and others well above 3, increasing the cross-sectional dispersion of kurtosis, driving AG rejections despite a lower mean. The increase in JB rejections is likely attributable primarily to rising skewness rather than excess kurtosis, suggesting that the distributional departure during the crisis is driven more by asymmetry than by heavy tails per se. This implies that Assumption 4 is not severely violated in either period, as the moment condition concerns tail thickness rather than skewness.

Additionally, the $n = 126$ rolling window results are broadly consistent with the main findings, confirming that our conclusions are not sensitive to the choice of rolling window length. The shorter window yields higher estimated edge densities in both periods, as expected, but the qualitative pattern of rising interconnectedness and the forecasting paradox are preserved across both specifications.

6 Conclusion

This paper makes two contributions to the empirical evaluation of FNETS. First, we provide the first systematic test of the core FNETS assumptions on empirical financial data, documenting how their validity changes across market regimes. Second, we estimate FNETS networks and forecasting performance across a calm period (2004–2006) and a crisis period (2007–2009), linking assumption violations to observed model behavior.

Our results reveals that the stationarity assumption fails almost entirely during the crisis period, reflecting extreme volatility persistence driven by the collapse of short-term debt backing markets. Estimated network density rises substantially during the crisis, consistent with the correlated deleveraging that followed the funding shock, though this increase should be interpreted with caution given the concurrent stationarity violation. Most counterintuitively, FE_{max} improves significantly during the crisis despite these assumption violations, pointing to a forecasting paradox that the current analysis cannot fully resolve.

Two directions for future work follow naturally. First, formally establishing the mechanism behind the forecasting paradox remains an open question. A plausible hypothesis is that the endogenous nature of the 2008 crisis which is rooted in the fragility of short-term debt markets and the high interconnectedness of financial institutions, may cause all 46 firms to move together in a sustained directional pattern, making one-step-ahead forecasts structurally easier. Testing this hypothesis against a contrasting episode would require an exogenous shock to the financial sector, that is, the COVID-19 pandemic. Unlike the 2008 crisis, which originated from within the financial system itself, COVID-19 represented an exogenous shock to the broader economy, with financial institutions acting as lenders of first resort rather than as the source of systemic fragility (Li et al., 2020). If the forecasting paradox is specific to endogenous crises where common financial sector exposure drives co-movement, it may not replicate during COVID. Conversely, if it does replicate, a more general mechanism may be at work. We leave this comparison to future work.

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9 Appendix

1. Figure 3 plots the cross-sectional mean of the log-volatility series using $\bar{X}_t = \frac{1}{46}\sum_{i=1}^{46} X_{it}$ across the calm (2004–2006) and crisis (2007–2009) sub-periods. Dashed horizontal lines indicate the period-specific means. During the calm period, $\bar{X}_t$ fluctuates in a narrow band around a stable low mean, providing visual support for the stationarity of the underlying series.

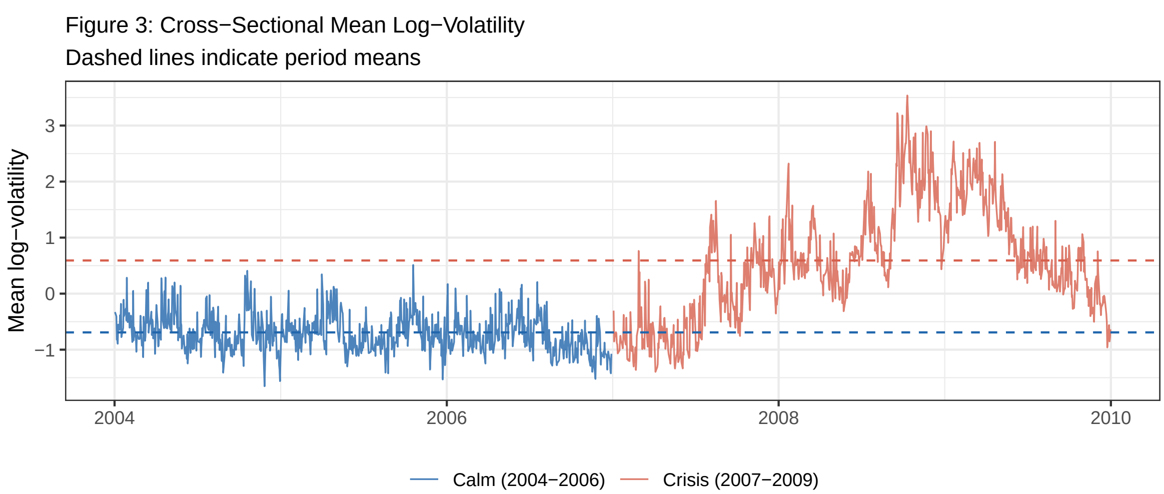

Figure 3: Cross-Sectional Mean Log-volatility

2. Figure 4 presents boxplots of FE_{avg} across the calm and crisis periods under both window specifications. The distributions of FE_{avg} are broadly similar across the two sub-periods under both specifications, consistent with the statistically insignificant t-test results reported in Table 4.

Both periods exhibit substantial right skew and a number of large outliers, reflecting occasional episodes of large average forecast errors.

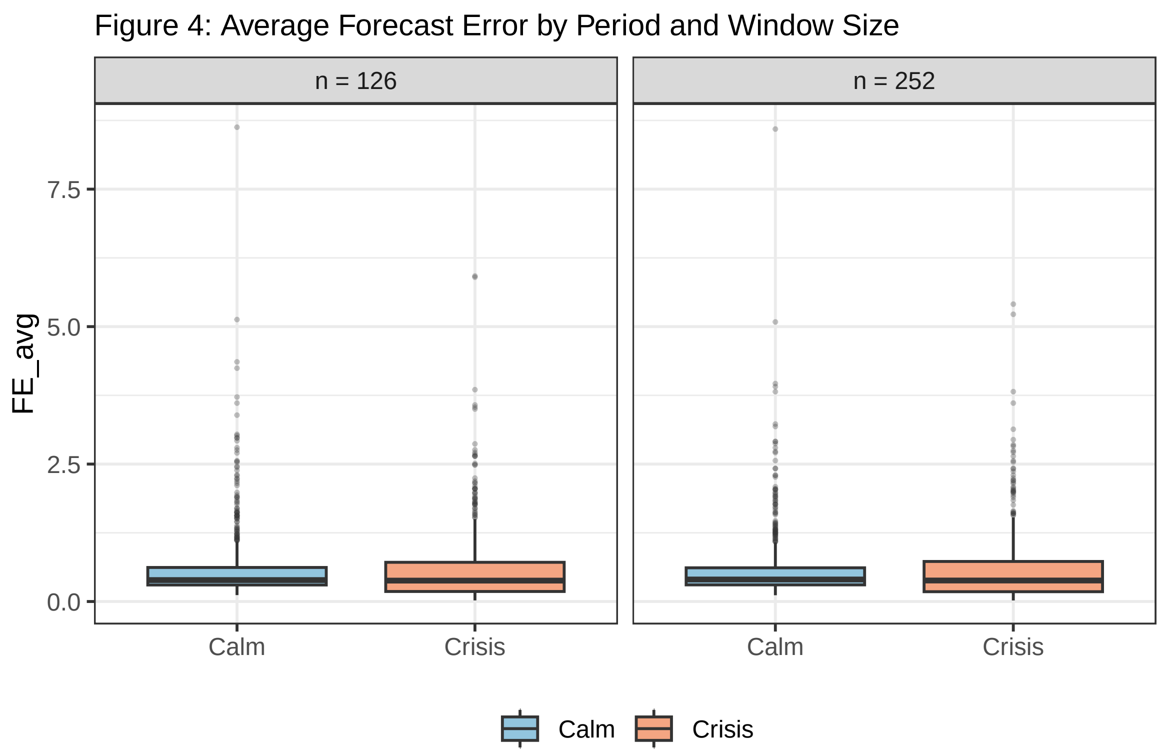

Figure 4: Average Forecast Error

3. Figure 5 presents box-plots of FE_{max} across the calm and crisis periods under both rolling window specifications (n = 252 and n = 126). Under both specifications, the crisis period exhibits a lower median and a tighter inter-quartile range compared to the calm period, consistent with the statistically significant difference in means reported in Table 4. The distribution of FE_{max} during the calm period is more dispersed, with a higher median and a longer upper tail.

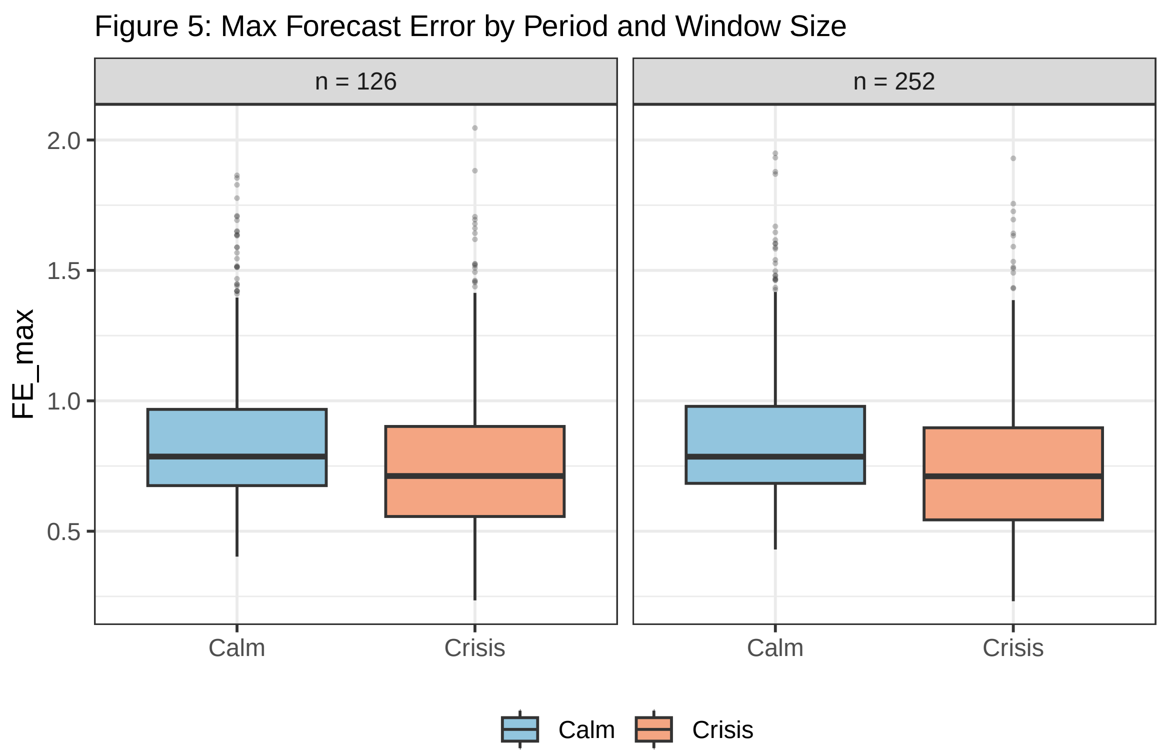

Figure 5: Max Forecast Error